11-Color / 3-Laser Flow Cytometry Protocol

DN B Cell Subset Gating (DN1 / DN2 / DN3)

For Extrafollicular B Cell Studies in Dengue and Comparative Immunology

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1. Rationale

Double-negative (DN) B cells (IgD- CD27- within CD19+) are a key population in extrafollicular B cell responses. They are expanded in active SLE (mean ~19% of CD19+ B cells vs. ~5% in healthy donors; Wei et al. 2007), severe COVID-19 (~19% of CD19+ B cells in ICU patients; Woodruff et al. 2020), and are hypothesised to be relevant in dengue infection.

This protocol uses an 11-color, 3-laser panel to identify the total DN compartment AND resolve the functionally distinct DN1, DN2, and DN3 subsets. The panel includes CD21 (FITC) and CD11c (PE) -- the two markers required to subdivide DN cells per the Jenks/Sanz classification (Jenks et al. 2018; Sanz 2025). CD38 (BV421) enables plasmablast exclusion, and CD24 (AF700) helps distinguish transitional B cells. A viability dye (eFluor506) and lineage dump channel (CD3/CD14 on BV711) ensure clean B cell gating.

2. Panel Design

2.1 Complete Antibody Panel

Laser	Fluorochrome	Marker	Role in DN Gating
Violet	BV421	CD38	PB exclusion; DN gate (CD38-)
Violet	eFluor506	L/D	Dead cell exclusion
Violet	BV711	CD3 / CD14	Lineage dump (T cells, monocytes)
Violet	BV785	IgD	CRITICAL: Naive/switched axis
Blue	FITC	CD21	DN1 vs DN2/DN3 resolution
Blue	PE	CD11c	DN2 vs DN3 resolution
Blue	PE-Cy7	CD66b	Granulocyte exclusion
Red	RB705	CD19	CRITICAL: Pan-B cell gate
Red	APC	CD27	CRITICAL: Memory axis
Red	AF700	CD24	Transitional B cell ID
Red	APC-Fire750	CD45	Leukocyte gate (whole blood)

Panel capability: This panel can fully resolve DN1, DN2, and DN3 subsets within the DN gate. The core DN gate uses IgD (BV785) and CD27 (APC); the DN subdivision uses CD21 (FITC) and CD11c (PE). CD38 (BV421) separates plasmablasts from switched memory. CD24 (AF700) identifies transitional B cells. CD45 (APC-Fire750) and CD66b (PE-Cy7) suggest this panel is designed for whole blood or minimally processed samples.

3. Spectral Overlap & Compensation Concerns

With 11 fluorochromes across 3 lasers, spectral overlap is the primary technical risk. The following analysis identifies high-risk pairs and provides mitigation strategies. Overlaps are ranked by severity: CRITICAL (may compromise gating), MODERATE (manageable with good compensation), and LOW (routine compensation).

3.1 Critical Overlaps

(A) RB705 (CD19) <-> AF700 (CD24) -- RED LASER

Both RB705 (~705 nm emission) and AF700 (~723 nm emission) are excited by the 633 nm red laser with heavily overlapping emission spectra. This is the most problematic pair in the panel. CD19 is highly expressed on all B cells, so even modest spillover of RB705 into the AF700 detector will create a false-positive CD24 signal on CD19-bright cells.

- Mitigation: Titrate CD19-RB705 to the minimum concentration that cleanly separates CD19+ from CD19-. Use single-stain RB705 beads to measure exact spillover into AF700 channel.
- Mitigation: Set the CD24 (AF700) gate using a strict FMO-AF700 control (all antibodies INCLUDING CD19-RB705 but excluding CD24-AF700). This FMO will show the spillover-induced baseline shift.
- Mitigation: If spillover is unacceptable, consider swapping CD19 to a different fluorochrome (e.g., BV605 if a 5th violet detector is available) or CD24 to BV510/BV605.
- Impact on DN gating: CD24 is used to exclude transitional B cells (CD24+) from the DN gate. If CD24 signal is unreliable due to RB705 spillover, transitional cells may contaminate the DN gate. This is especially problematic in dengue, where transitional B cells can be expanded.

(B) APC (CD27) <-> AF700 (CD24) <-> APC-Fire750 (CD45) -- RED LASER CASCADE

Three fluorochromes on the red laser in adjacent detectors creates a compensation cascade: APC (~660 nm) spills into AF700 (~723 nm), and AF700 spills into APC-Fire750 (~787 nm). Over-subtraction in one pair can create artefactual negative populations in the next.

- Mitigation: Compensate APC -> AF700 first, then AF700 -> APC-Fire750. Verify that CD27-bright cells (plasmablasts) do not show artefactual CD24 positivity.
- Mitigation: Use single-colour controls on CELLS (not just beads) for APC, since beads may not capture the dynamic range of CD27 expression on plasmablasts (CD27-hi).
- Impact on DN gating: CD27 is a CRITICAL axis for DN identification (CD27-). Any spillover from APC into AF700 is tolerable as long as the CD27- boundary is set correctly by FMO. The reverse (AF700 spilling into APC) is lower risk because CD24 is not brightly expressed.

RED LASER CONGESTION: 4 fluorochromes (RB705, APC, AF700, APC-Fire750) on a single laser is the maximum advisable for conventional cytometry. Compensation will be complex. Run ALL four single-stain controls at the beginning of each experiment. If compensation cannot be resolved, prioritise CD19 and CD27 (essential) over CD24 and CD45 (supportive).

3.2 Moderate Overlaps

(C) BV421 (CD38) <-> eFluor506 (L/D) -- VIOLET LASER

BV421 has a broad emission tail extending past 500 nm, overlapping with the eFluor506 detector (~506 nm). CD38 is expressed at varying levels on all B cells (bright on plasmablasts, dim on memory). Spillover will shift the viability baseline for CD38-bright cells.

- Mitigation: Use BV421 single-stain beads to measure spillover into the eFluor506 channel. Set the live/dead gate using FMO for eFluor506 that INCLUDES CD38-BV421.
- Impact: If dead cells with high autofluorescence overlap with CD38-bright plasmablasts, the viability gate may

exclude live plasmablasts. Draw the L/D gate generously for CD38-hi cells.

(D) BV711 (CD3/CD14) <-> BV785 (IgD) -- VIOLET LASER

BV711 (~711 nm) can spill into the BV785 detector (~785 nm). CD3 and CD14 are brightly expressed on T cells and monocytes respectively. Since these are dump-channel markers on cells that should be excluded BEFORE the IgD gate is applied, spillover into BV785 is only a concern if the dump gate leaks.

- Mitigation: Apply the CD3/CD14 dump gate BEFORE IgD vs CD27 plotting. Cells that pass the dump gate (CD3- CD14-) will have minimal BV711 signal and thus minimal spillover into BV785.
- Impact: Low risk if gating order is correct. Do NOT plot IgD vs CD27 on ungated cells.

(E) PE (CD11c) <-> PE-Cy7 (CD66b) -- BLUE/YG LASER

PE-Cy7 tandem dyes are susceptible to degradation, releasing unconjugated PE that emits at ~575 nm and appears as false PE signal. If CD66b-PE-Cy7 degrades, granulocytes will appear falsely CD11c-PE+.

- Mitigation: Protect PE-Cy7 conjugates from light. Aliquot and freeze; do not use expired lots. Check lot-to-lot variation in PE-Cy7 tandem integrity.
- Mitigation: Since CD66b is a granulocyte marker and granulocytes should be excluded by scatter gating and/or CD45 gating, tandem breakdown is only an issue if granulocytes persist in the B cell gate. Verify that the lymphocyte scatter gate excludes granulocytes completely.
- Impact on DN gating: CD11c (PE) is CRITICAL for DN2 vs DN3 resolution. False PE signal on residual granulocytes could be misinterpreted as CD11c+ DN2 cells. Always confirm that CD11c+ events in the DN gate are CD66b-.

3.3 Low-Risk Overlaps

- FITC (CD21) -> PE (CD11c): Standard spillover, well-handled by compensation. Both markers are critical for DN subdivision -- verify with FMO.
- BV785 (IgD) -> APC-Fire750 (CD45): Different lasers (violet vs red), so cross-laser excitation is minimal. No action needed beyond standard compensation.
- eFluor506 (L/D) -> BV711 (CD3/CD14): Minor violet laser spillover. Manageable because dead cells are excluded before lineage gating.

3.4 Compensation Controls Checklist

Control	Fluorochrome	Notes
Comp beads + CD19	RB705	CRITICAL: match to antibody lot
Comp beads + CD27	APC	Use cells if beads underperform
Comp beads + CD24	AF700	Verify vs RB705 spillover
Comp beads + CD45	APC-Fire750	Check tandem integrity
Comp beads + IgD	BV785	Verify vs BV711 spillover
Comp beads + CD38	BV421	Measure spillover into eFluor506
Comp beads + CD3 or CD14	BV711	Use brightest marker for comp
Comp beads + L/D	eFluor506	Use amine-reactive on beads
Comp beads + CD21	FITC	Standard
Comp beads + CD11c	PE	Standard
Comp beads + CD66b	PE-Cy7	Check tandem breakdown

FMO controls required (minimum): FMO-CD27 (APC), FMO-IgD (BV785), FMO-CD11c (PE), FMO-CD21 (FITC), FMO-CD24

(AF700), FMO-CD38 (BV421). These six FMOs define the critical gate boundaries. The CD27 and IgD FMOs are essential -- they define the DN quadrant. The CD11c and CD21 FMOs are essential -- they define DN1/DN2/DN3.

4. Sample Preparation

4.1 Sample Type

The inclusion of CD45-APC-Fire750 and CD66b-PE-Cy7 suggests this panel is designed for whole blood or lysed whole blood. Both workflows are described below.

Option A: Whole blood (lyse-no-wash or lyse-wash)

- Collect 2-5 mL peripheral blood in EDTA or heparin tubes.
- Aliquot 100 uL whole blood per staining tube.
- Add surface antibody cocktail. Incubate 20-30 min at RT in the dark.
- Add 2 mL 1x RBC lysis buffer (e.g., BD FACS Lysing Solution). Incubate 10 min at RT.
- Centrifuge 300g x 5 min, aspirate supernatant.
- Wash 1x with PBS + 2% FBS. Resuspend in 200-300 uL for acquisition.
- Note: L/D eFluor506 is an amine-reactive dye -- add BEFORE fixation/lysis. For lyse-no-wash protocols, the L/D dye must be added to whole blood before the antibody cocktail.

Option B: Ficoll-isolated PBMCs

- Isolate PBMCs by Ficoll-Hypaque density gradient (Anolik et al. 2004; Tipton et al. 2015).
- CD45 gating is unnecessary (all PBMCs are CD45+) -- the APC-Fire750 channel becomes available for other purposes or serves as a quality control gate.
- CD66b is unnecessary (granulocytes removed by Ficoll) -- PE-Cy7 channel is freed.
- Stain 0.5-1.0 x 10^6 PBMCs per tube. 100 uL staining buffer, 20-30 min at 4C in the dark.
- Wash 2x with PBS + 2% FBS, resuspend in 200-300 uL for acquisition.

L/D staining order: eFluor506 (amine-reactive viability dye) must be added in protein-free buffer (PBS without FBS) BEFORE antibody staining. FBS contains free amines that quench amine-reactive dyes. Incubate L/D 15-20 min at RT in PBS, then wash once, then add antibody cocktail in staining buffer.

5. Gating Strategy

The gating hierarchy below follows the Jenks/Sanz classification (Jenks et al. 2018; Sanz 2025; Woodruff et al. 2020 Table 1) adapted to this specific panel.

5.1 Hierarchical Gating (Steps 1-8)

```
Step 1: FSC-A vs SSC-A       -> Leukocyte gate (exclude debris)
Step 2: FSC-A vs FSC-H       -> Singlet gate (exclude doublets)
Step 3: eFluor506 vs FSC-A   -> Live gate (eFluor506-negative = live)
Step 4: CD45 vs SSC-A        -> Lymphocyte gate (CD45+ SSC-low)
        [Skip if using PBMCs]
Step 5: CD3/CD14 vs FSC-A    -> Dump gate (CD3- CD14- = non-T, non-mono)
        CD66b vs FSC-A       -> Granulocyte exclusion (CD66b-)
        [Skip CD66b if using PBMCs]
Step 6: CD19 vs SSC-A        -> B cell gate (CD19+)

Step 7: IgD vs CD27          -> Four-quadrant plot (within CD19+):

              CD27+
           +-----------+-----------+
           |           |           |
           |   USM     |    SM     |
           | IgD+CD27+ | IgD-CD27+|
           |           |  (+ PB)  |
```

```
           +-----------+-----------+
           |           |           |
           |  Naive    |    DN     |
           | IgD+CD27- | IgD-CD27-| <- TARGET GATE
           |           |           |
           +-----------+-----------+
              IgD+         IgD-

Step 8: CD21 vs CD11c        -> DN SUBSET RESOLUTION (within DN gate):

              CD11c+
           +-----------+-----------+
           |           |           |
           |  (rare)   |   DN2     |
           | CD21+     | CD21-     |
           | CD11c+    | CD11c+    | <- EF EFFECTOR
           +-----------+-----------+
           |           |           |
           |   DN1     |   DN3     |
           | CD21+     | CD21-     |
           | CD11c-    | CD11c-    | <- PRE-PB
           +-----------+-----------+
              CD21+        CD21-
```

5.2 DN Subset Definitions

Subset	CD21 (FITC)	CD11c (PE)	Biology	Expected in HD
DN1	+	-	GC-like memory precursor	Dominant (~60-70% of DN)
DN2	-	+	EF pre-plasmablast (T-bet+)	Minor (~10-20% of DN)
DN3	-	-	Pre-PB (T-bet-)	Minor (~10-20% of DN)

In active disease (SLE, severe COVID-19), DN2 becomes the dominant fraction (~80% of DN), DN1 contracts, and DN3 expands moderately. The DN2:DN1 ratio is the most robust single metric of EF pathway activation (Woodruff et al. 2020).

5.3 Additional Gates Enabled by This Panel

Plasmablast gate (within IgD- CD27+ quadrant):

- CD38-hi (BV421) + CD27-hi (APC) = plasmablasts
- Further subdivide by CD19 level: CD19-hi = early PB; CD19-lo = maturing PB

Transitional B cell exclusion (within CD19+ gate):

- CD24+ (AF700) + CD38-hi (BV421) + IgD+ = transitional B cells
- Exclude from DN gate: transitional B cells are CD38-hi and will not fall in the DN gate (which is CD38-low/neg by definition), but verify with CD24 overlay.

Activated naive (aNAV) identification (within Naive quadrant):

- IgD+ CD27- CD21-lo (FITC) CD11c+ (PE) = activated naive B cells
- These are DN2 precursors in the EF pathway (Jenks et al. 2018). Same CD21/CD11c logic as DN subdivision but applied within the naive (IgD+CD27-) quadrant.

6. Critical Gating Notes & Pitfalls

6.1 IgD (BV785) Boundary

- IgD expression is bimodal. The FMO-IgD control sets the boundary between IgD+ and IgD-.
- BV785 is a tandem dye: verify lot-to-lot consistency and protect from light to prevent degradation. Degraded BV785 releases unconjugated BV421 fluorescence, which will spill into the CD38 (BV421) channel -- creating false CD38 positivity on IgD-bright naive cells.
- If BV785 tandem integrity is compromised, IgD-bright cells may appear CD38-intermediate, confounding the transitional B cell and plasmablast gates.

6.2 CD27 (APC) Boundary

- CD27 has a continuous expression spectrum (negative -> dim -> bright). The FMO-CD27 control is ESSENTIAL for setting the CD27- boundary. Do not use isotype controls.
- CD27-bright events are plasmablasts (within IgD- gate). Ensure the CD27+ gate captures both dim (memory) and bright (PB) populations.
- RB705 (CD19) spillover into APC (CD27) is minimal (~660 vs ~705 nm); not a concern.

6.3 CD11c (PE) Boundary for DN2 vs DN3

- CD11c expression on DN2 cells is typically moderate (not as bright as on monocytes/DCs). The FMO-PE control must be run within the DN gate context -- spillover from FITC (CD21) into PE can shift the baseline.
- Residual monocytes that escape the CD14 dump gate will be CD11c-bright. Overlay CD14 (BV711) on the CD11c vs CD21 plot to confirm that CD11c+ events in the DN gate are CD14-.
- PE-Cy7 (CD66b) tandem breakdown: verify that CD11c+ events in the DN gate are CD66b- to exclude granulocyte contamination (whole blood only).

6.4 CD21 (FITC) Boundary for DN1 vs DN2/DN3

- CD21 expression on B cells ranges from bright (resting naive, DN1) to negative (activated populations, DN2, DN3). The boundary between CD21+ and CD21- is less bimodal than IgD -- use FMO-FITC control carefully.
- FITC has low brightness. Ensure adequate antibody titre for CD21 separation. Under-titration will compress the CD21+ population toward the negative boundary.

GATING ORDER MATTERS: Always apply gates in this sequence: Singlets -> Live -> Dump(CD3/CD14/CD66b) -> CD19+ -> IgD vs CD27 -> DN subset. Applying the IgD vs CD27 plot before removing T cells and monocytes will show contaminating populations that mimic B cell subsets.

7. Expected Results & Quality Checks

7.1 Expected Frequencies (% of CD19+ B cells)

Population	Healthy Donors	Active SLE	Severe COVID-19
Naive	55-75%	Variable	Variable
USM	5-15%	Often reduced	Reduced (CoV-A)
SM	15-25%	Variable	Variable
Total DN	3-7% (mean ~5%)	10-40%	~19% (ICU)

7.2 DN Subset Composition (% of DN cells)

Subset	Healthy Donors	Active SLE	Severe COVID-19
DN1	60-70%	Contracted	~10% (ICU)

DN2	10-20%	Dominant (>50%)	~80% (ICU)
DN3	10-20%	Expanded	Expanded

Reference values: Wei et al. 2007 (n=29 HD, n=36 SLE); Jenks et al. 2018 (DN subset %); Woodruff et al. 2020 (n=17 HD, n=10 ICU-C, n=7 SLE).

7.3 Key Derived Metrics

- DN2:DN1 ratio: log2(DN2/DN1). In HD: negative (DN1 > DN2). In active SLE and severe COVID-19: positive (DN2 > DN1). This ratio is the most informative single metric of EF pathway activation (Woodruff et al. 2020; P <= 0.0001 vs HD).
- Total DN frequency (% of CD19+): >10% in a healthy donor is unusual -- verify gating.
- aNAV frequency (% of naive B cells): Elevated aNAV within the IgD+CD27- quadrant (CD21-lo CD11c+) indicates active EF pathway engagement.

7.4 Quality Control Checkpoints

- CD19+ B cells: 5-15% of lymphocytes (healthy PB). <2% suggests sample issue or depletion.
- L/D: >85% viability expected for fresh samples. <70% compromises all downstream gates.
- Dump channel: CD3+ and CD14+ events should be <5% of CD19+ gate (T-B doublets, monocyte contamination).
- CD66b+: Should be absent from lymphocyte gate if scatter gating is correct (whole blood).
- IgD bimodality: Clear separation of IgD+ and IgD- peaks. Unimodal distribution suggests staining failure.

8. Dengue-Specific Considerations

No published study has yet characterised DN1/DN2/DN3 subsets in dengue using the Jenks/Sanz classification. This panel enables that characterisation for the first time. The following considerations are extrapolated from comparative immunology data:

- Acute dengue (days 3-7 post-fever-onset) produces a massive plasmablast expansion (up to 50% of CD19+ B cells). Use CD38 (BV421) vs CD27 (APC) to quantify PBs separately from SM in the IgD-CD27+ quadrant. PBs may partially obscure DN expansion if only total DN is reported.
- DENV is a single-stranded RNA virus -- TLR7 is the relevant innate sensor. DN2 cells are hyper-responsive to TLR7 (Jenks et al. 2018). If DN2 expansion occurs in dengue, it should be detectable with this panel's CD21 vs CD11c resolution.
- Secondary dengue infection (prior heterotypic serotype exposure) may show a mixed DN profile: both recalled DN memory cells (potentially DN1-like) and de novo EF-derived DN2 cells. The DN2:DN1 ratio may distinguish primary (predominantly DN2) from secondary (mixed) responses.
- Severity stratification is critical: record DF vs DHF/DSS (or WHO 2009 categories) and days post-fever-onset. By analogy with COVID-19, DN2 expansion may correlate with severity (Woodruff et al. 2020: DN2% correlated with log(CRP), r2=0.39).
- CD24 (AF700): Dengue can induce transitional B cell expansion. CD24+CD38+ transitional cells should be identified and excluded from DN analyses.
- Cross-reactivity context: Bank serum for DENV serotype-specific ELISA/ELISpot. Correlate DN2 frequency with cross-reactive vs type-specific antibody output -- this is the key question for understanding whether EF responses contribute to ADE-relevant antibodies.

9. Data Reporting Recommendations

When reporting results from this protocol, include:

- Total DN frequency (% of CD19+ B cells) and absolute count if available.
- DN subset composition: DN1, DN2, DN3 as % of total DN cells.
- DN2:DN1 ratio (log2-transformed) as the primary EF activation metric.
- aNAV frequency (% of naive B cells) as an upstream EF pathway indicator.
- Plasmablast frequency (CD38-hi CD27-hi within IgD- gate) reported separately from SM.
- All four IgD/CD27 quadrant frequencies to contextualise the DN result.
- Total CD19+ B cell count (absolute) -- frequency shifts can reflect expansion of other compartments rather than true DN expansion.
- Days post-fever-onset, primary vs secondary infection, severity classification.
- FMO control plots for CD27, IgD, CD21, and CD11c alongside sample plots.
- Compensation matrix or spillover values for the red laser channels (RB705/APC/AF700/APC-Fire750).

10. Panel Optimisation Notes

10.1 If Red Laser Compensation Fails

If the RB705 (CD19) <-> AF700 (CD24) spillover proves unresolvable:

- Option 1: Drop CD24-AF700 from the panel entirely. Transitional B cell exclusion can be achieved by CD38-hi + IgD+ gating alone (less precise but functional).

- Option 2: Move CD19 to a violet channel (e.g., BV605 if detector available) to eliminate the red laser congestion problem.
- Option 3: Move CD24 to PE-Cy5 or PerCP-Cy5.5 (blue laser) if those detectors are free. This eliminates the red laser conflict entirely.

10.2 If PBMCs Are Used Instead of Whole Blood

Two channels become redundant and can be repurposed:

- CD45 (APC-Fire750): All PBMCs are CD45+. Repurpose this channel for CXCR5 (confirms DN1/DN2 subdivision independently of CD21) or FCRL5 (DN2 marker).
- CD66b (PE-Cy7): No granulocytes after Ficoll. Repurpose for IgG or IgM subclass analysis, or for a second lineage exclusion marker (e.g., CD56 for NK cell dump).

CXCR5 is the single most discriminating marker between DN1 and DN2 (Jenks et al. 2018). If working with PBMCs, strongly consider adding CXCR5 on the freed APC-Fire750 channel. CXCR5 + CD21 together provide redundant DN1/DN2 resolution and allow cross-validation of the CD21-based gating strategy.

11. References (Wiki Sources)

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